AI454 — Natural Language Processing || Assignment 2

Team Members:

RAMA JARADAT

JANA ASHOUR

Part 1: LSTM vs BiLSTM

Trial	Hidden Size	Dropout	Optimizer	Learning Rate	Batch Size	Epochs
T1	64	0.2	Adam	1e-3	32	15
T2	128	0.3	Adam	5e-4	32	15
T3	256	0.5	Adam	1e-4	64	20
T4	256	0.3	SGD	0.1	32	25

T1

Test result

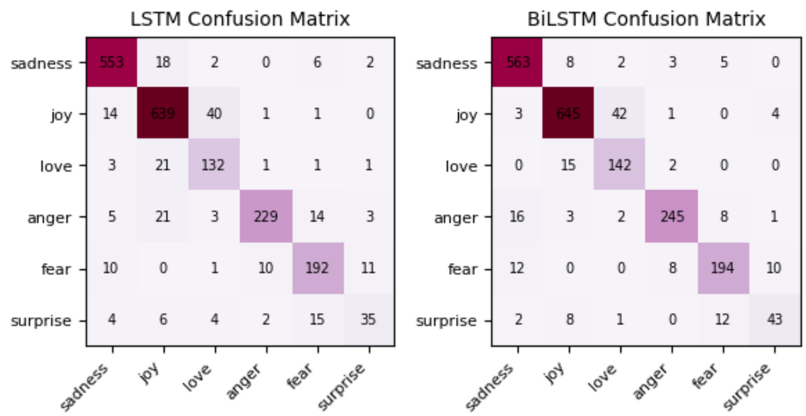
Model	Dataset	Accuracy	Macro Precision	Macro Recall	Macro F1
LSTM	Dev	**0.8985**	0.8694	0.8569	0.8621
	Test	**0.8900**	0.8374	0.8203	0.8262
BiLSTM	Dev	**0.9245**	0.9014	0.8985	0.8989
	Test	**0.9160**	0.8698	0.8664	0.8664

LSTM Train Acc: 0.9551 , BiLSTM Train Acc: 0.9959

Per class

Emotion	LSTM Precision	BiLSTM Precision	LSTM Recall	BiLSTM Recall	LSTM F1	BiLSTM F1
sadness	0.9446	0.9363	0.9690	0.9618	0.9567	0.9489
joy	0.9499	0.9591	0.9281	0.9318	0.9389	0.9452

Emotion	LSTM Precision	BiLSTM Precision	LSTM Recall	BiLSTM Recall	LSTM F1	BiLSTM F1
love	0.7513	0.8010	0.8931	0.9045	0.8161	0.8496
anger	0.9459	0.9430	0.8909	0.9018	0.9176	0.9219
fear	0.8858	0.8925	0.8661	0.9009	0.8758	0.8967
surprise	0.7414	0.8767	0.6515	0.7901	0.6935	0.8312



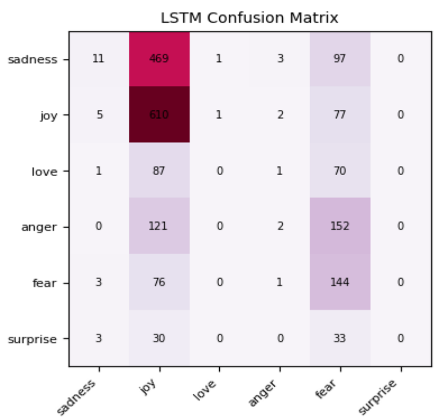
T2

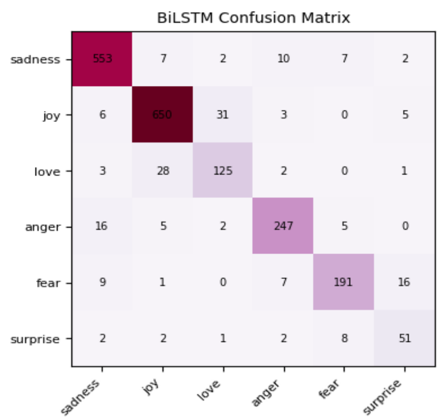
Model	Dataset	Accuracy	Macro Precision	Macro Recall	Macro F1
LSTM	Dev	0.3810	0.1412	0.2537	0.1577
	Test	0.3835	0.2316	0.2578	0.1660
BiLSTM	Dev	0.9180	0.8886	0.8864	0.8871
	Test	0.9085	0.8583	0.8661	0.8616

Per class

Emotion	LSTM Precision	LSTM Recall	LSTM F1	BiLSTM Precision	BiLSTM Recall	BiLSTM F1
sadness	0.4783	0.0189	0.0364	0.9389	0.9518	0.9453
joy	0.4379	0.8777	0.5843	0.9380	0.9353	0.9366
love	0.0000	0.0000	0.0000	0.7764	0.7862	0.7812

Emotion	LSTM Precision	LSTM Recall	LSTM F1	BiLSTM Precision	BiLSTM Recall	BiLSTM F1
anger	0.2222	0.0073	0.0141	0.9114	0.8982	0.9048
fear	0.2513	0.6429	0.3614	0.9052	0.8527	0.8782
surprise	0.0000	0.0000	0.0000	0.6800	0.7727	0.7234





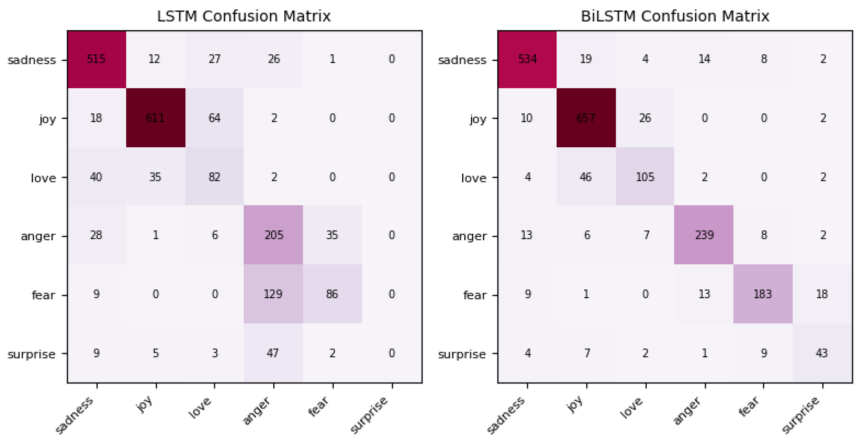
T3

Model	Dataset	Accuracy	Macro Precision	Macro Recall	Macro F1
LSTM	Dev	0.7510	0.5696	0.5856	0.5717
	Test	0.7495	0.5658	0.5684	0.5551
BiLSTM	Dev	0.8850	0.8468	0.8364	0.8405
	Test	0.8805	0.8239	0.8104	0.8165

Per class

Emotion	LSTM Precision	BiLSTM Precision	LSTM Recall	BiLSTM Recall	LSTM F1	BiLSTM F1
sadness	0.8320	0.9303	0.8864	0.9191	0.8583	0.9247
joy	0.9202	0.8927	0.8791	0.9453	0.8992	0.9182
love	0.4505	0.7292	0.5157	0.6604	0.4809	0.6931
anger	0.4988	0.8885	0.7455	0.8691	0.5977	0.8787
fear	0.6935	0.8798	0.3839	0.8170	0.4943	0.8472

Emotion	LSTM Precision	BiLSTM Precision	LSTM Recall	BiLSTM Recall	LSTM F1	BiLSTM F1
surprise	0.0000	0.6232	0.0000	0.6515	0.0000	0.6370

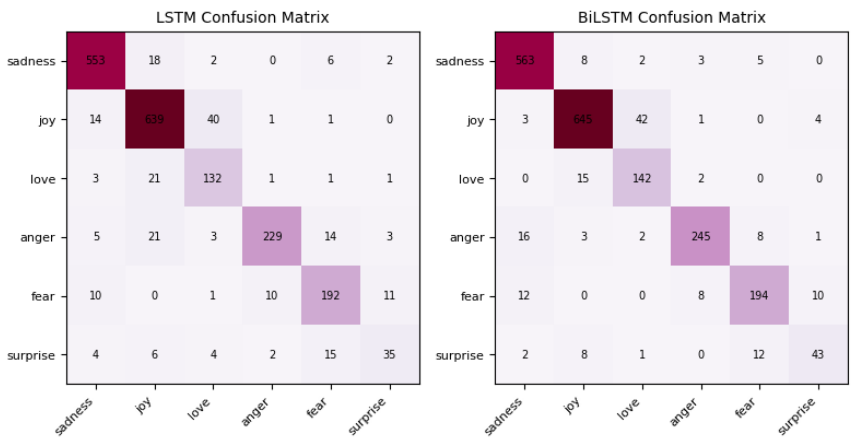


T4

Model	Dataset	Accuracy	Macro Precision	Macro Recall	Macro F1
LSTM	Dev	0.3490	0.0977	0.1662	0.0939
	Test	0.3470	0.1135	0.1677	0.0986
BiLSTM	Dev	0.9240	0.9188	0.8813	0.8971
	Test	0.9160	0.8983	0.8513	0.8702

Per class

Emotion	LSTM Precision	BiLSTM Precision	LSTM Recall	BiLSTM Recall	LSTM F1	BiLSTM F1
sadness	0.3333	0.9468	0.0465	0.9501	0.0816	0.9485
joy	0.3476	0.9144	0.9597	0.9525	0.5103	0.9331
love	0.0000	0.9040	0.0000	0.7107	0.0000	0.7958
anger	0.0000	0.9197	0.0000	0.9164	0.0000	0.9180
fear	0.0000	0.8648	0.0000	0.9420	0.0000	0.9017
surprise	0.0000	0.8400	0.0000	0.6364	0.0000	0.7241



The best combination of hyperparameters found for each model :

Trial one for both :

Trial	Hidden Size	Dropout	Optimizer	Learning Rate	Batch Size	Epochs
T1	64	0.2	Adam	1e-3	32	15

LSTM vs BiLSTM

Metric	LSTM	BiLSTM	Difference
Parameters	1,564,286	1,607,166	+42,880 (+2.7%)
Avg. Time/Epoch	~2.5s	~2.8s	+0.3s (+12%)
Best Dev F1	0.7510	0.8983	+0.1473

Part 2: BiLSTM vs BiRNN

reuse the best BiLSTM implementation from Part 1 without modification

Trial	Hidden Size	Dropout	Optimizer	Learning Rate	Batch Size	Epochs
T1	64	0.2	Adam	1e-3	32	15

Test results

Model	Dataset	Accuracy	Macro Precision	Macro Recall	Macro F1
BiLSTM	Dev	0.9245	0.9014	0.8985	0.8989
BiLSTM	Test	0.9160	0.8698	0.8664	0.8664
BiRNN	Dev	0.7310	0.5060	0.5060	0.4798
BiRNN	Test	0.7450	0.4606	0.5061	0.4803

BiRNN Train Acc: 0.4936 , BiLSTM Train Acc: 0.9959

Per class

Emotion	BiLSTM Precision	BiRNN Precision	BiLSTM Recall	BiRNN Recall	BiLSTM F1	BiRNN F1
sadness	0.9446	0.9219	0.9690	0.9139	0.9567	0.9179
joy	0.9499	0.7580	0.9281	0.9554	0.9389	0.8453
love	0.7513	0.0000	0.8931	0.0000	0.8161	0.0000
anger	0.9459	0.5355	0.8909	0.6582	0.9176	0.5905
fear	0.8858	0.5481	0.8661	0.5089	0.8758	0.5278
surprise	0.7414	0.0000	0.6515	0.0000	0.6935	0.0000

BiLSTM vs BiRNN

Metric	BiLSTM	BiRNN	Difference
Parameters	1,607,166	1,707,774	+100,608 (+6.3%)
Avg. Time/Epoch	~2.8s	~2.7s	-0.1s
Best Dev F1	0.8983	0.7672	-0.1311

1. Accuracy and F1 Score

- **BiLSTM significantly outperforms BiRNN across all metrics**
- **BiLSTM learns context in both directions more effectively**
- **BiRNN shows difficulty with minority classes (love, surprise) → F1 = 0**

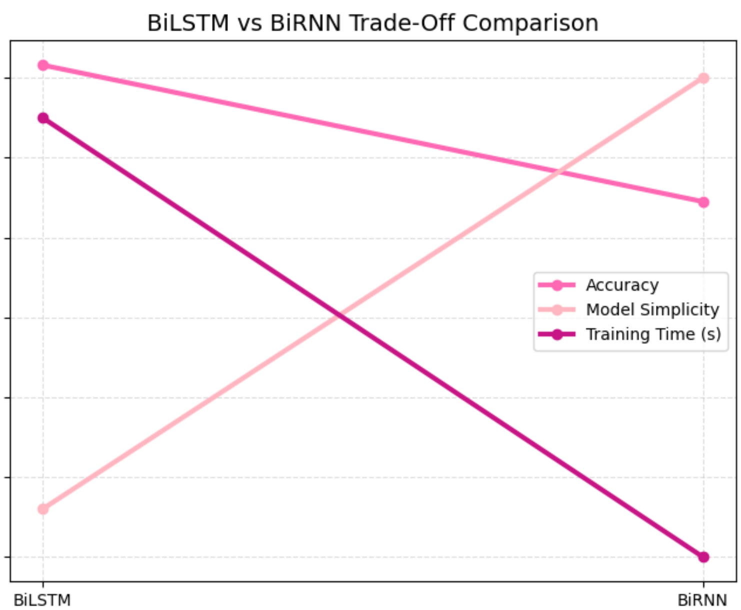
2. Generalization

- **BiLSTM Dev and Test results are very close, indicating strong generalization**
- **BiRNN has bigger Dev vs Test difference → less stable learning**

3. Per-Class Performance

- **BiLSTM handles all classes well, especially small classes**
- **BiRNN collapses on low-frequency classes, predicting the majority instead**

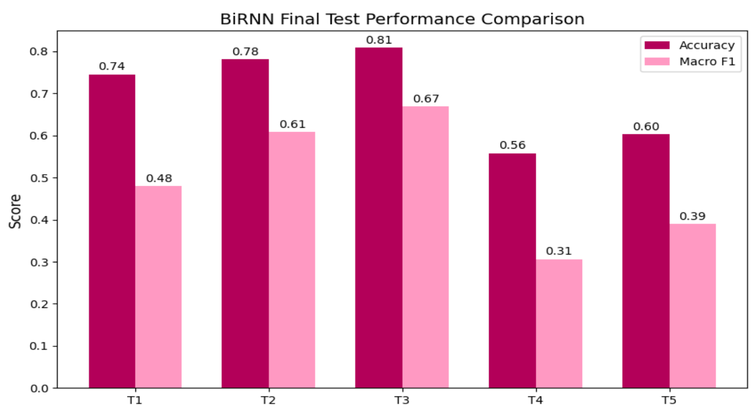
4.Trade-offs in terms of accuracy, model simplicity, and training time



A summary of the hyperparameters tuned specifically for BiRNN.:

Trial	Hidden Size	Dropout	Optimizer	Learning Rate	Batch Size	Epochs
T1	64	0.2	Adam	1e-3	32	15
T2	128	0.3	Adam	5e-4	32	15
T3	256	0.5	Adam	1e-4	64	20
T4	256	0.3	SGD	1e-2	32	25
T5	512	0.4	Adam	7e-4	64	25

Results : **Highest results is T3 -Train Acc: 0.8799-Dev Accuracy: 0.8000**

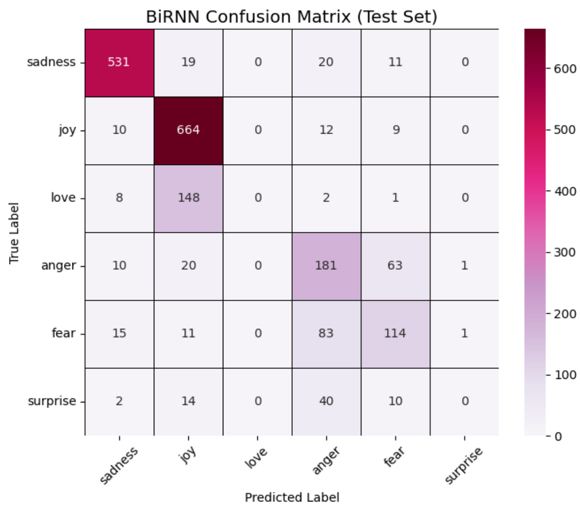


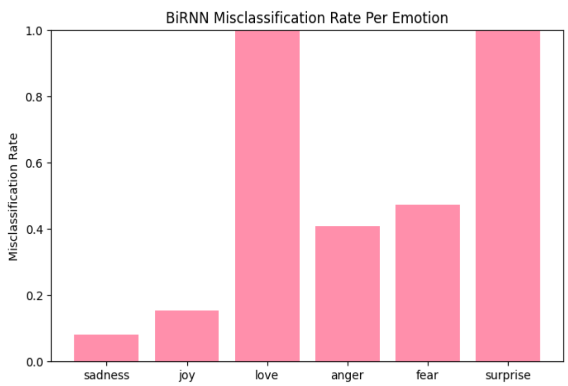
1. **Limitations RNNs**

- **Suffering from vanishing gradients, making it hard to learn long-range dependencies.**
- **RNNs underperform on subtle emotional categories, as reflected in confusion matrix patterns**

2. **Key Insights From Confusion Matrices**

- **Strong performance on high-frequency and explicit emotions like sadness and joy.**
- **Very weak performance on rare classes (love, surprise) — almost always misclassified.**
- **Misclassifications appear mainly between semantically similar emotions (e.g., anger ↔ sadness, fear ↔ anger).**
- **Many predictions cluster along majority classes → indicates bias due to imbalance.**





Part 3: Effect of Additive Attention on BiLSTM

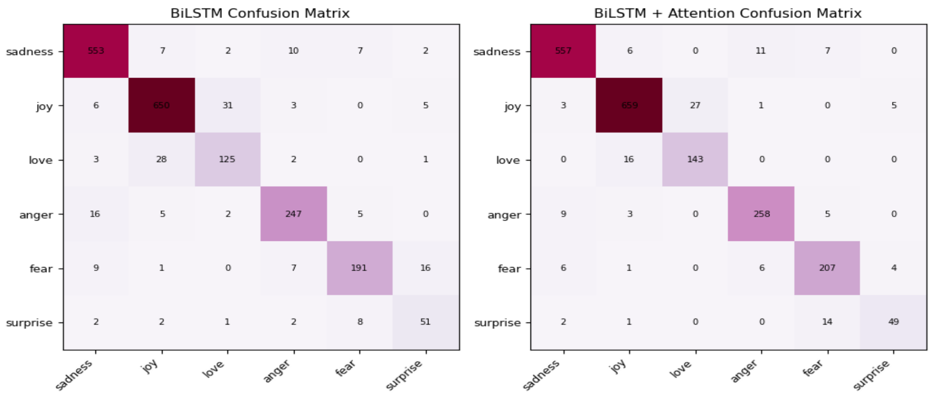
- **BiLSTM vs BiLSTM with Additive Attention**

  Attention Dim = 128 for BiLSTM with Additive Attention

Trial	Hidden Size	Dropout	Optimizer	Learning Rate	Batch Size	Epochs
T1	64	0.2	Adam	1e-3	32	15

- Performance Comparison:

Model	Train Accuracy	Train F1	Dev Accuracy	Dev F1 (Best)	Test Accuracy	Test F1
BiLSTM	0.9959	0.9920	0.9225	0.8951	0.9175	0.8802
BiLSTM + Attention	0.9906	0.9856	0.9325	0.9107	0.9295	0.8885



- hyperparameter tuning summary with evaluation and comparison

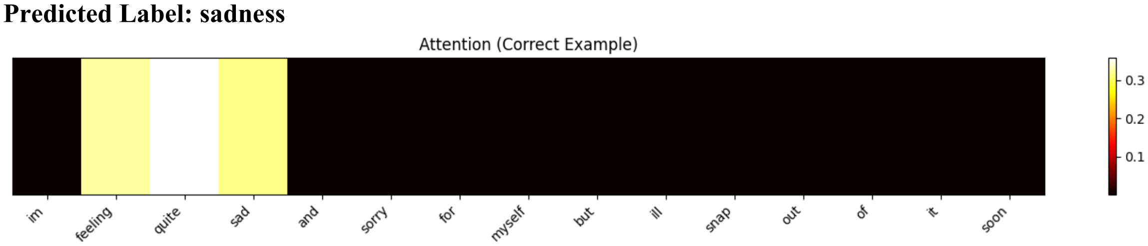
Trial	Hidden Size	Dropout	Optimizer	Learning Rate	Batch Size	Epochs	Attention Dim	Test-f1	Dev-f1
T1	64	0.2	Adam	1e-3	32	15	128	0.9031	0.9209
T2	512	0.3	Adam	0.01	32	40	256	0.8519	0.8814

BiLSTM vs BiLSTM + Attention

Metric	BiLSTM	BiLSTM + Attention	Difference
Parameters	1,607,166	1,623,806	+16,640 (+1.0%)
Avg. Time/Epoch	~2.8s	~3.1s	+0.3s (+11%)
Best Dev F1	0.8983	0.9135	+0.0152

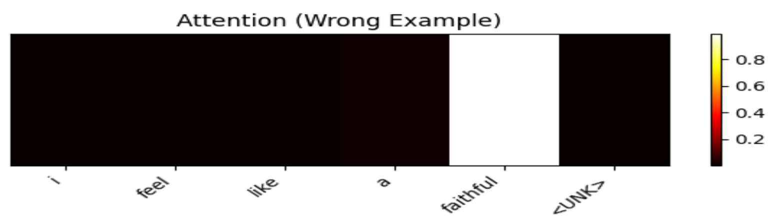
- **Attention visualization for:**

1. Correct Example
True Label : sadness
Predicted Label: sadness

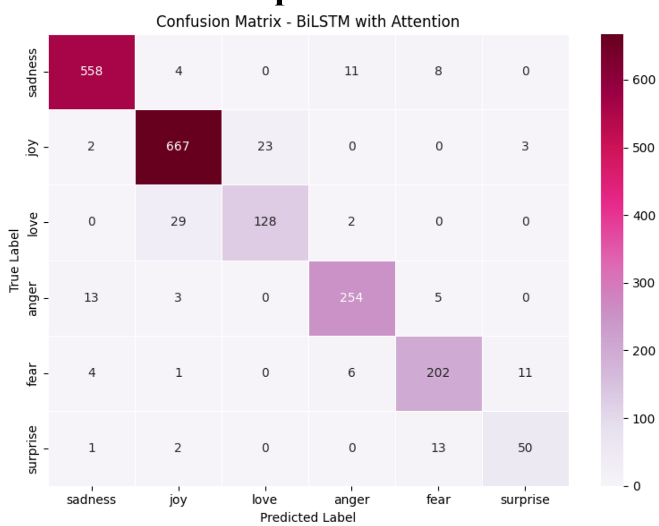


2. Wrong Example
True Label : love

Predicted Label: joy



- **patterns observed in correct vs incorrect examples according to the heatmap:**



Incorrect Predictions

- **Confusion occurs between semantically similar emotions (e.g., *love vs joy*, *fear vs surprise*).**

Correct Predictions

- **Attention focuses on strong emotional keywords (e.g., *sad*, *happy*, *angry*).**

- **Tokens the model tends to focus on for different emotions: The attention mechanism prioritizes emotionally strong and sentiment-bearing keywords, enabling more accurate emotion recognition.**

Part 4: Self-Attention Transformer Encoder with Learned Positional Encodings

T1

- **Dropout rate: 0.1**
- **Hidden size (Feed-forward dimension): 256**
- **Number of attention heads: 4**
- **Number of Transformer encoder layers: 2**
- **Learning rate: 5e-4**
- **Optimizer: Adam**
- **Number of epochs: 20**
- **Batch size: 32**

Train Accuracy	Dev Accuracy	Test Accuracy	Train Macro F1	Dev Macro F1	Test Macro F1	Test Macro Precision	Test Macro Recall
0.9964	**0.9120**	**0.8910**	**0.9943**	**0.8861**	**0.8320**	**0.8382**	**0.8265**

Parameters 1,711,918

Avg Time / Epoch (sec) 3.44

T2

The same hyperparameters for the best bilstm model

- **Hidden size (feed-forward layer): 64**
- **Number of attention heads: 4**
- **Number of encoder layers: 2**
- **Dropout rate: 0.2**
- **Learning rate: 1e-3**
- **Optimizer: Adam**
- **Number of epochs: 15**
- **Batch size: 32**

Model	**Dataset**	**Accuracy**	**Macro Precision**	**Macro Recall**	**Macro F1**	**Training Accuracy**
BiLSTM (No Attention)	Dev	0.9245	0.9014	0.8985	0.8989	**0.9959**

Model	Dataset	Accuracy	Macro Precision	Macro Recall	Macro F1	Training Accuracy
	Test	0.9160	0.8698	0.8664	0.8664	**0.9959**
Transformer Encoder	Dev	0.9135	0.8913	0.8899	**0.8903**	**0.9928**
	Test	0.9005	0.8489	0.8525	**0.8495**	**0.9928**

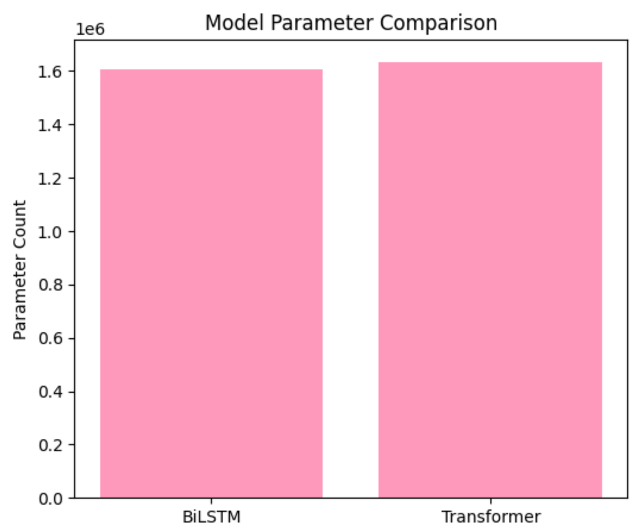
the Hyperparameters in t1 for transformer overperform all the models

Self-attention improves performance when emotional cues are distributed across the sentence or when the sentence is longer. It is especially effective for emotions like joy, anger, and sadness that rely on contextual relationships. However, for short or subtle emotions (like surprise or love), BiLSTM performs better, likely due to simpler structure and stronger generalization on limited data.

- **Parameter Count & Training Time Comparison Table**

BiLSTM vs Transformer

Metric	BiLSTM	Transformer	Difference
Parameters	1,607,166	1,634,734	+27,568 (+1.7%)
Avg. Time/Epoch	~2.8s	~5.4s	+2.6s (+93%)
Best Dev F1	0.8983	0.8973	-0.0010



Model Comparison Table

Model	Trainable Parameters	Avg. Training Time per Epoch
LSTM (unidirectional)	1,564,286	~2.5s
BiLSTM	1,607,166	~2.8s
BiRNN	1,707,774	~2.7s
BiLSTM + Attention	1,623,806	~3.1s
Transformer	1,634,734	~5.4s